# thermal-bond RCF

## PROCESS GUIDE

thermal-bond 3.0F (VT-4B3H RCF)

UL Approval: E214381 Version: 20/09/2026

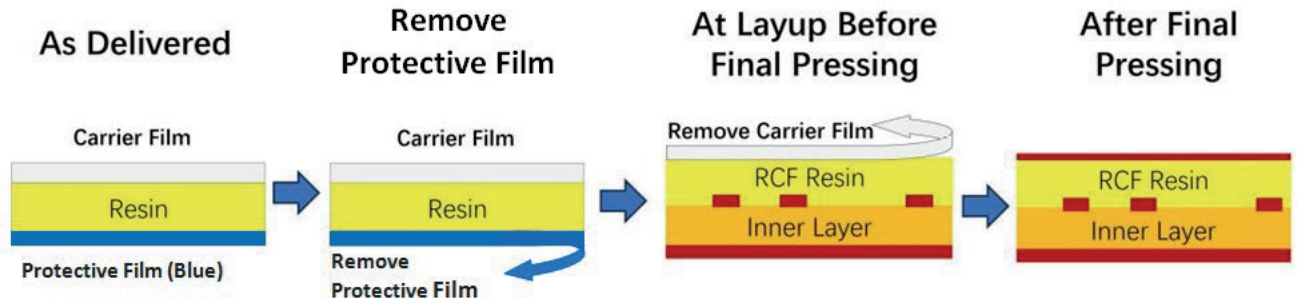
## Precautions in Handling

- Storage Condition & Shelf Life

| | | RCF | |
|---|---|---|---|
| Storage Condition | Temperature | Below 23°C (73°F) | Below 5°C (41°F) |
| | Relative Humidity | Below 55% | / |
| Shelf Life | | 3 Months | 6 Months |

- The thermal management prepreg should be taken care of when handling and wearing rubber gloves to prevent contamination is strongly recommended during processing.

## Product Handling & Press Process Flow




1. Please put RCF on a clean and flat surface with the PET film side up in cleanroom.
2. Please pay more attention to the alignment of RCF colloid edge with the substrate or metal plate.
3. Start from one corner when peeling off the PET film from resin.
4. Then put copper foil or another piece of RCF on the surface.
5. Avoid RCF sticking resulting from stacking during pre-stacking of multiple PCBs or prolonged standing time.
6. Unused RCF shall be vacuum-sealed and stored in cold storage. Allow temperature acclimatization before reuse.

Disclaimer:

> The information and data contained in this technical literature is based on data and knowledge correct at the time of publishing/ printing and is believed to be accurate and is offered in good faith for the benefit of the user. The user should make his own tests to verify the suitability of this product for any application before its use. All data are typical values only and subject to change without notice.

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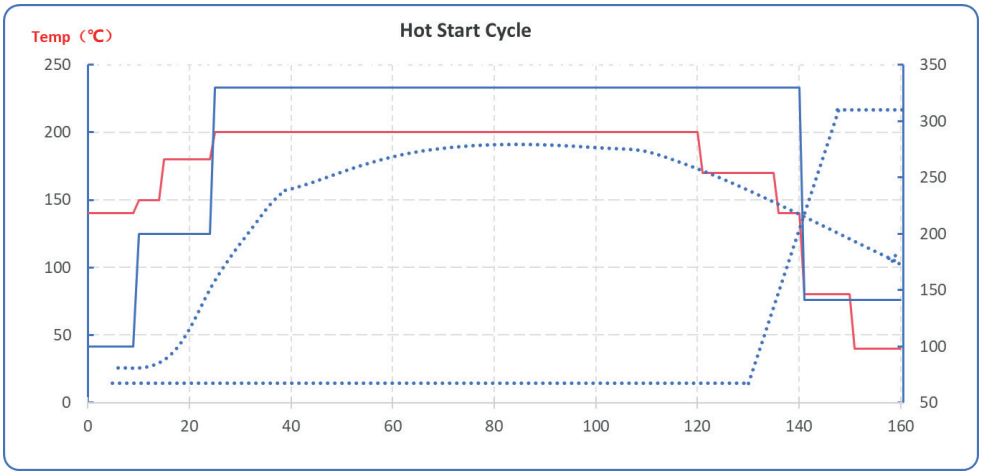
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## Press Parameters

1. Heating rate (Rate of Rise) of material [Material Temperature]: 1.5~3.5°C/min (2.7~5.4°F/min).
2. Curing Temperature & Time: >60min at more than 180°C (356°F) [Material Temperature].
3. Full Pressure: ≥300psi, increase properly if Aluminum ≥2mm
   It's a guideline that full pressure is started between 70~100°C of material temperature.
4. Vacuuming should be continued until over 140°C (284°F) [Material Temperature].
5. Cold Press condition: Keep Plate @ Room Temperature by water; Pressure: 100psi; Keep Time: 60minutes

| Product Name | Product Code | Full Pressure (psi) |
|---|---|---|
| VT-4B3H RCF | FT-R | ≥450 |
| VT-4B3H RCF | FT-M | ≥350 |
| VT-4B3H RCF | FT-H | ≥300 |




## Baking Recommendation During PCB Process

| Process | Purpose | Cycle | Potential Risk |
|---|---|---|---|
| Before HASL, PCB should be separate (not stacked) and supported in a rack for this process. HASL process should be finished within 24 hours after baking finishing. | To eliminate moisture | 2 hours @ 125°C | Potential for measling, blister and de-lamination. |
| Before shipment, if PCB stored for >1 month, PCB should be baked before packaging. | To eliminate moisture | 2~4 hours @ 125°C | Potential for measling, blister and de-lamination. |

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## Typical Drilling Condition

### Without Metal Base (only for reference)

| Diameter (mm) | Stack Height | Spindle Speed (KRPM) | Feed Rate (mm/s) | Retract Rate (mm/s) | Hit Count |
|---|---|---|---|---|---|
| 0.15~0.30 | 1 PNL/stack | 125~155 | 17~33 | 200~250 | 1200 |
| 0.35~0.40 | 1 PNL/stack | 95~110 | 40~47 | 250 | 1200 |
| 0.55~1.00 | 1 PNL/stack | 60~88 | 47 | 330 | 1200 |
| >1.00 | 1 PNL/stack | 50~70 | 30~47 | 300 | 800~1200 |

- Carbide drill bit is prone to excessive wear, preferably Jinzhou SHD/MDC series drill bit
- Aluminum and Phenolic Board Covers are recommended

### With Metal Base (only for reference)

| Diameter (mm) | Stack Height | Spindle Speed (KRPM) | Feed Rate (mm/s) | Retract Rate (mm/s) | Hit Count |
|---|---|---|---|---|---|
| 0.25~0.30 | 1 PNL/stack | 40 | 25~30 | 100~200 | 800 |
| 1.0~2.5 | 1 PNL/stack | 36~39 | 35~50 | 250~450 | 200~500 |
| >2.5 | 1 PNL/stack | 30~37 | 30~40 | 200~300 | 200~500 |

- Carbide drill bit is prone to excessive wear, preferably Jinzhou SHD/MDC series drill bit.
- Recommend Aluminum and Phenolic Board Covers

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## Image Transfer

| Process | Recommendation |
|---|---|
| Dry Film | The lamination speed maybe needs slower than standard FR4 to have metal substrate approach proper lamination temperature. |
| Wet Film | Both aluminum and copper could be coated to protect aluminum during etching process. Follow instruction details of manufacturers |
| Developing | Masking or film protection on aluminum is better during developing. Follow chemical manufacturer's recommendation. |

- Dry film and wet film are both applicable process on aluminum base laminate.
- Before filming, cleaning of panel surface is necessary.
- Wet film process option
  - Curtain
  - Screen

## Etching / Stripping

| Process | Recommendation |
|---|---|
| Etching | Alkaline or acid solution (cupric chloride, ferric chloride) are both applicable.<br>Alkaline etching is faster than acid etching and is fit for below 3oz copper. The acid etching performs well on preventing of undercut and over etch. |
| Stripping | Apply sodium hydroxide solution to remove etch resist and conveying process is preferred. |

## Solder Masking

| Process | Recommendation |
|---|---|
| Solder masking | Control the thickness between 1~2mil or depends on boards design |
| Pre-heating | Follow manufacturer's recommendation on the setting of temperature and time. Having a filter system in oven is preferred. |
| Developing | Solution concentration, temperature and spray pressure need to be controlled. |
| Curing | Follow the details of temperature and time of manufacturer. |

- Roughness of board surface is necessary to get better adhesion between boards and solder mask.
- Both thermal cured and UV cured solder mask are applicable.
- Masking options
  - Screen Printing
  - Curtain Coating
  - Spray Coating
  - Roller Coating

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## Surface Finish Options

| Process | Recommendation |
|---|---|
| HASL | The most popular surface treatment method. Peel off the protective film from aluminum before HASL. |
| OSP | Follow details of manufacturer's recommendation. |
| ENIG | Popular in wire bonding application. Follow chemistry recommendation to avoid defects of black pad and brickle gold. |
| Silver Immersion | Silver immersion has a better soldering ability but has disadvantage of silver migration. |

## Routing

| Diameter (mm) | Spindle Speed (KRPM) | Feed Rate (IPM) | Retract Rate (IPM) | Travel Speed (mm/sec) |
|---|---|---|---|---|
| 0.8 | 37 | 10~15 | 30 | 12~15 |
| 1.0 | 37 | 10~15 | 30 | 10~12 |
| 1.2 | 37 | 12~18 | 30 | 8~10 |
| 1.5 | 36 | 15~20 | 30 | 5~8 |
| 2.0 | 34 | 15~20 | 30 | 3~5 |

- Backup board – Phenolic is preferred.
- Tools material – Tungsten or Diamond carbite with 2 flutes is preferred
- Stack height – 1 panel / stack
- Consult your tool supplier for more advice.

## Punching

- Alloy side is upwards when punching.
- Tonnage – 120T and above is preferred.
- "D1&D2" should be greater than board thickness.
- "D3" should be greater than board thickness + 0.5mm.
- Consult your tool supplier for more advice.

